

Integrated Action, Kinetic Bound, and the Weighted Metric Trace Lemma (Route δ , Plan 2, Building Block WEIGHTEDMETRICTRACE)

Abstract

This building block chains three ingredients into a pointwise metric distance bound at a radius $\widehat{\rho}$ near $\rho_\delta = 1 - \kappa\sqrt{\delta_T}$. The starting point is the integrated quadratic Fusco–Julin oscillation (B^2 -int) of [4], which replaces any attempt at a pointwise L^2 radial trace by its $d\mu$ -averaged form. Combining (B^2 -int) with the per- ρ Cauchy deficit bound of [2] yields the integrated action bound $\int (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu \leq C \delta_T$. A Talenti / bad-set decomposition then divides by $-t'(\rho)$ on the good set and uses the elementary uniform L^1 -Lipschitz bound (A0) on the bad set, giving the kinetic bound (K) in Lebesgue form $\int |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C' \delta_T$. The abstract weighted metric trace lemma then produces $\widehat{\rho} \in [\rho_\delta - C_* \delta, \rho_\delta]$ with $d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \leq C_* \delta_T$. The chain uses no pointwise (R)/(1.2) anywhere.

1 Statements

Fix $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$, $\kappa > 0$. Let $\Omega \subset B_R$ open with $|\Omega| = \omega_n$, $u = u_\Omega$ the variational torsion function, $\delta := \delta_T(\Omega) \leq \delta_0(n, R, \rho_*)$. Set $E_\rho := \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$, $\rho_\delta := 1 - \kappa\sqrt{\delta}$, $\mu(d\rho) := (-t'(\rho))d\rho$. Define $F_\rho := \rho^{-1}E_\rho \subset B_{R/\rho_*}$, $\gamma(\rho) := [F_\rho] \in \mathcal{X}$.

Theorem 1.1 (Integrated action bound). *Conditional on (F) of [3] and (B^2 -int) of [4]:*

$$\int_{\rho_*}^{\rho_\delta} (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu(\rho) \leq C(n, R, \rho_*, \kappa) \delta_T(\Omega). \quad (1)$$

Theorem 1.2 (Kinetic bound (K), Lebesgue form).

$$\int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C'(n, R, \rho_*, \kappa) \delta_T(\Omega). \quad (2)$$

Theorem 1.3 (Weighted metric trace). *There exist C_* and $\widehat{\rho} \in [\rho_\delta - C_* \delta, \rho_\delta]$ such that $d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \leq C_* \delta_T(\Omega)$.*

2 The integrated action bound (5.2-int)

From [2] Theorem (T), $|\dot{F}_\rho|_{\mathcal{X}} \leq C \inf_a \int |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho|$; squaring gives $|\dot{F}_\rho|_{\mathcal{X}}^2 \leq 2[(\int |V_\rho - \bar{V}_\rho|)^2 + (\int |H_{z_\rho, \rho} - \bar{V}_\rho|)^2]$.

D_H -piece. The unconditional Cauchy–Schwarz on V_ρ (algebraic from [1]): $(\int |V - \bar{V}|)^2 \leq C D_H$. Integrating: $\int (\int |V - \bar{V}|)^2 d\mu \leq C \int D_H d\mu \leq C \delta$.

Radial-trace piece. Using $|H - \bar{V}| \leq ||x - z|/\rho - 1| + |1 - \bar{V}|$ plus Cauchy–Schwarz on $\partial^* E_\rho$: $\int (\int |H - \bar{V}|)^2 d\mu \leq C \delta$ via (B^2 -int) of [4], plus perimeter excess integration and the slice-rearrangement $|E \Delta B_\rho|^2 \leq C \Pi + (G3)$.

Distance piece. $d_{\mathcal{X}}(F_{\rho}, B_1) \leq \rho^{-n} |E_{\rho} \Delta B_{\rho}(z_{\rho})|$ pointwise, so $\int d_{\mathcal{X}}^2 d\mu \leq C \int |E_{\rho} \Delta B_{\rho}|^2 d\mu \leq C\delta$ via the slice-rearrangement.

Combining gives (1). $\square$

Remark 2.1. The (T1²-int) input from [4] replaces the conditional pointwise (R)/(1.2). The chain is rate-preserving thanks to the slicing-then-squaring trick.

3 The kinetic bound (K)

Lemma 3.1 (Uniform metric Lipschitz, (A0)). $|\dot{F}_{\rho}|_{\mathcal{X}} \leq L_0(n, R, \rho_*)$ a.e.

Proof. Nestedness gives $|E_{\rho'} \Delta E_{\rho}| \leq n\omega_n |\rho' - \rho|$. Split $F_{\rho} \Delta F_{\rho'}$ through $(\rho')^{-1} E_{\rho}$ and use the dilation estimate to get $d_{\mathcal{X}}([F_{\rho'}], [F_{\rho}]) \leq L_0 |\rho' - \rho|$. $\square$

Lemma 3.2 (Bad-set Lemma). $|B_{\tau_0}| := |\{\rho : -t'(\rho) < \tau_0 = \rho_*/(2n)\}| \leq K_2\delta$.

Proof. On B_{τ_0} , $|\psi(\rho)| = -t'_B - (-t') \geq \rho_*/(2n)$. By the integrated Talenti gap $\int |\psi| d\rho \leq K_1\delta$, $(\rho_*/(2n))|B_{\tau_0}| \leq K_1\delta$. $\square$

Proof of Theorem 1.2. Split $[\rho_*, \rho_{\delta}] = G \sqcup B_{\tau_0}$ with G the good set where $-t' \geq \tau_0$. On G : $\int_G |\dot{F}_{\rho}|^2 d\rho \leq \tau_0^{-1} \int_G |\dot{F}_{\rho}|^2 (-t') d\rho \leq (2n/\rho_*) \cdot C\delta$. On B_{τ_0} : by (A0), $\int_{B_{\tau_0}} |\dot{F}_{\rho}|^2 d\rho \leq L_0^2 K_2\delta$. Adding gives (2). $\square$

4 The weighted metric trace lemma

Theorem 4.1 (Abstract weighted metric trace). Assume (H1) $\int (d_{\mathcal{X}}^2 + |\dot{\gamma}|^2) d\mu \leq C\delta$, (H2) $\mu([a, b]) \geq c(b - a) - C\delta$, (H3) $\mu_{\rho} \leq M$, (5.14) $\int |\dot{\gamma}|^2 d\rho \leq C'\delta$. Then there is $\hat{\rho} \in [\rho_{\delta} - C_*\delta, \rho_{\delta}]$ with $d_{\mathcal{X}}(\gamma(\hat{\rho}), B_1)^2 \leq C_*\delta$.

Proof. Step 1: macroscopic Markov on a fixed sub-interval. Fix $L^* := (\rho_{\delta} - \rho_*)/4$, $J^* := [\rho_{\delta} - L^*, \rho_{\delta} - L^*/2]$. By (H2), $\mu(J^*) \geq cL^*/4$. By (H1) + Markov on J^* , there is $\rho_0 \in J^*$ with $d_{\mathcal{X}}(\gamma(\rho_0), B_1)^2 \leq K_1\delta$ where $K_1 = 4C/(cL^*)$.

Step 2: kinetic transport. For any $\hat{\rho} \in I_{\delta} := [\rho_{\delta} - K_2\delta, \rho_{\delta}]$, AGS Thm 1.1.2 + Cauchy-Schwarz: $d_{\mathcal{X}}(\gamma(\rho_0), \gamma(\hat{\rho}))^2 \leq (\hat{\rho} - \rho_0) \int |\dot{\gamma}|^2 d\rho \leq L^* \cdot C'\delta$ by (5.14).

Step 3: triangle. $d_{\mathcal{X}}(\gamma(\hat{\rho}), B_1)^2 \leq 2K_1\delta + 2L^*C'\delta =: C_*\delta$. $\square$

Verification of hypotheses for the torsion application:

- (H1) = Theorem 1.1.
- (H2) interval bound from Talenti / Nicola-Tilli profile gap [1].
- (H3) $-t' \leq 1/n$ via Talenti.
- (5.14) = Theorem 1.2.

Proof of Theorem 1.3. Apply Theorem 4.1 to $\gamma = [F_{\rho}]$. $\square$

5 Conditionality

- Theorem 1.1: consumes (B²-int) + (G3) of [4], unconditional given (F) of [3].
- Theorem 1.2: plus elementary (A0) and bad-set Lemma, both unconditional.
- Theorem 1.3: plus (H2), (H3) from Talenti, unconditional.

The whole block is unconditional given (F). Pointwise (R)/(1.2) is never invoked.

References

- [1] Building block LEVELSETIDENTITY.
- [2] Building block METRICFRAMEWORK.
- [3] Building block CENTROIDBOUND.
- [4] Building block SPACETIMEIDENTITY.
- [5] Building block BOUNDARYLAYERTRANSFER.
- [6] L. Ambrosio, N. Gigli, G. Savaré, *Gradient Flows in Metric Spaces*, Birkhäuser, 2008. Thm 1.1.2.
- [7] G. Talenti, Ann. Mat. Pura Appl. (4) **110** (1976), 353–372.
- [8] F. Maggi, Cambridge University Press, 2012.